Alberto Santini

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Current affiliation
Universitat Pompeu Fabra, Spain
Associate professor and "Ramón y Cajal" researcher at the Department of Economics and Business.
Affiliate professor at the Centre for Studies on Planetary Wellbeing.

Past Positions

2023–2024, 2017–2021 — Universitat Pompeu Fabra, Spain
Tenure-track assistant professor and "Juan de la Cierva" researcher at the Department of Economics and Business.

2021–2023 — ESSEC Business School, France
Marie Skłodowska-Curie Fellow at the Department of Information Systems, Decision Sciences and Statistics.

2017 — RWTH Aachen University, Germany
Postdoctoral Researcher at the Deutsche Post Chair for the Optimisation of Distribution Networks.
Chair: Prof. Michael Schneider.

Other affiliations

2023–now — Global Challenges Lab, RWTH Aachen University, Germany (Partner)

2018–now — Barcelona School of Management, Spain (Professor)

2018–now — Barcelona Graduate School of Mathematics, Spain (Affiliate professor)

2018–now — Barcelona School of Economics, Spain (Affiliate professor)

2017–2023 — RWTH Aachen University Business School, Germany (Lecturer)

Habilitations

2023 — Certificado R3
Certificate in the Spanish higher education system. It recognises researchers who have reached the European Union's R3 profile "Established Researcher". Success rate in the 2023 call: 1/3.

2023 — Acreditació de Recerca (Professor Agregat)
Habilitation as Associate Professor in the Catalan higher education system.

2022 — Abilitazione Scientifica Nazionale a Professore di Seconda Fascia 01/A6
Habilitation as Associate Professor in the Italian higher education system.

Education

2014–2016 University of Bologna, Italy
Ph.D. in Automatic Control and Operational Research.
Supervisors: Silvano Martello, Daniele Vigo.

2011–2013 University of Copenhagen, Denmark
M.Sc. in Mathematics, part of the Master of Excellence programme.
Thesis advisors: Stefan Røpke and Louise Kallehague. Mark: 12/12.

2008–2010 University of Catania, Italy
B.Sc. in Mathematics.
Thesis advisor: Angelo Bella. Mark: 110/110 summa cum laude.

Publications Journal papers

2026

1. Alberto Santini and Vikrant Vaze (2026). "Integrated Schedule Planning for Regional Airlines Using Column Generation". Accepted, to appear in: **Transportation Science**
2. Rossana Cavagnini, Alberto Santini, Michael Schneider, and Murwan Siddig (2026). "Recent Developments in Location Routing Problems: Deterministic multi-period, multi-echelon, and multi-objective problems". In: **European Journal of Operational Research** 334 (3), pp. 705–719. DOI: 10.1016/j.ejor.2026.04.001
3. Rossana Cavagnini, Alberto Santini, and Michael Schneider (2026). "Recent Developments in Location Routing Problems: Deterministic single-echelon, single-objective, single-period problems". In: **European Journal of Operational Research** 332 (3), pp. 711–729. DOI: 10.1016/j.ejor.2025.09.040
4. Enrico Malaguti, Paolo Paronuzzi, and Alberto Santini (2026). "Algorithms and Complexity Results for the 0–1 Knapsack Problem with Group Fairness". In: **Optimization Letters** 20, pp. 577–594. DOI: 10.1007/s11590-025-02252-y

2025

5. Diego Delle Donne, Alberto Santini, and Claudia Archetti (2025). "Integrating Public Transport in Sustainable Last-Mile Delivery: Column Generation Approaches". In: **European Journal of Operational Research** 324 (1), pp. 75–91. DOI: 10.1016/j.ejor.2024.12.047
6. Minakshi Punam Mandal, Alberto Santini, and Claudia Archetti (2025). "Tactical workforce sizing and scheduling decisions for last-mile delivery". In: **European Journal of Operational Research** 323 (1), pp. 153–169. DOI: 10.1016/j.ejor.2024.12.006
7. Alberto Santini (2025). "Destination selection and flight scheduling for regional airlines at slot-constrained airports". In: **International Transactions in Operational Research** 32 (3), pp. 1400–1421. DOI: 10.1111/itor.13505

2024

8. Alberto Santini and Enrico Malaguti (2024). "The min-Knapsack Problem with Compactness Constraints and Applications in Statistics". In: **European Journal of Operational Research** 312 (1), pp. 385–397. DOI: 10.1016/j.ejor.2023.07.020

2023

9. Alberto Santini, Michael Schneider, Thibaut Vidal, and Daniele Vigo (2023). "Decomposition strategies for vehicle routing heuristics". In: **INFORMS Journal on Computing** 35.3, pp. 543–559. DOI: 10.1287/ijoc.2023.1288
10. Alberto Santini and Claudia Archetti (2023). "The hazardous orienteering problem". In: **Networks** 81 (2), pp. 235–252. DOI: 10.1002/net.22129

2022

11. Michele Monaci, Ciara Pike-Burke, and Alberto Santini (2022). "Exact algorithms for the 0–1 time-bomb knapsack problem". In: **Computers & Operations Research** 145. Article ID 105848. DOI: 10.1016/j.cor.2022.105848
12. Maxence Delorme and Alberto Santini (2022). "Energy-efficient automated vertical farms". In:

Omega 109. Article ID 102611. DOI: 10.1016/j.omega.2022.102611

13. Alberto Santini, Ana Viana, Xenia Klimentova, and João Pedro Pedroso (2022). "The probabilistic Travelling Salesman Problem with crowdsourcing". In: **Computers & Operations Research** 142. Article ID 105722. DOI: 10.1016/j.cor.2022.105722

2021

14. Christian Blum, Marko Djukanovic, Alberto Santini, Hua Jiang, Chu-Min Li, Felip Manyà, and Günter Raidl (2021). "Solving common longest subsequence problems via a transformation to the maximum clique problem". In: **Computers & Operations Research** 125. Article ID 105089. DOI: 10.1016/j.cor.2020.105089

15. Alberto Santini, Michael Schneider, Enrico Bartolini, and Vinicius Greco de Lemos (2021). "The crop growth planning problem in vertical farming". In: **European Journal of Operational Research** 294 (1), pp. 377–390. DOI: 10.1016/j.ejor.2021.01.034

16. Alberto Santini (2021). "Optimising the assignment of swabs and reagents for PCR testing during a viral epidemic". In: **Omega** 102. Article ID 102341. DOI: 10.1016/j.omega.2020.102341

2020

17. Renata Turkeš, Kenneth Sörensen, Lars Magnus Hvattum, Eva Barrena, Hayet Chentli, Leandro Coelho, Iman Dayarian, Axel Grimault, Anders Gullhav, Çağatay Iris, Merve Keskin, Alexander Kiefer, Richard Martin Lusby, Geraldo Regis Mauri, Marcela Monroy-Licht, Sophie Parragh, Juan-Pablo Riquelme-Rodríguez, Alberto Santini, Vínicius Gandra Martins Santos, and Charles Thomas (2020). "Data for a meta-analysis of the adaptive layer in adaptive large neighborhood search". In: **Data in Brief** 33. DOI: 10.1016/j.dib.2020.106568

2019

18. Alberto Santini (2019). "An Adaptive Large Neighbourhood Search algorithm for the Orienteering Problem". In: **Expert Systems with Applications** 123, pp. 154–167. DOI: 10.1016/j.eswa.2018.12.050

19. Denis Cornaz, Fabio Furini, Enrico Malaguti, and Alberto Santini (2019). "A note on selective line-graphs and partition colorings". In: **Operations Research Letters** 47 (6), pp. 565–568. DOI: 10.1016/j.orl.2019.08.005

2018

20. Enrico Malaguti, Silvano Martello, and Alberto Santini (2018). "The Traveling Salesman Problem with Pickups, Deliveries, and Draft Limits". In: **Omega** 74, pp. 50–58. DOI: 10.1016/j.omega.2017.01.005

21. Fabio Furini, Enrico Malaguti, and Alberto Santini (2018). "An Exact Algorithm for the Partition Coloring Problem". In: **Computers & Operations Research** 92, pp. 170–181. DOI: 10.1016/j.cor.2017.12.019

22. Alberto Santini, Stefan Ropke, and Lars Magnus Hvattum (2018). "A comparison of acceptance criteria for the Adaptive Large Neighbourhood Search metaheuristic". In: **Journal of Heuristics** 24 (5), pp. 783–815. DOI: 10.1007/s10732-018-9377-x

2017

23. Alberto Santini, Stefan Ropke, and Christian E.M. Plum (2017). "A branch-and-price approach to the Feeder Network Design Problem". In: **European Journal of Operational Research** 264 (2), pp. 607–622. DOI: 10.1016/j.ejor.2017.06.063

24. Andrea Bettinelli, Alberto Santini, and Daniele Vigo (2017). "A real-time conflict solution algorithm for the Train Rescheduling Problem". In: **Transportation Research Part B: Methodological** 106, pp. 237–265. DOI: 10.1016/j.trb.2017.10.005

2015

25. Alberto Santini, Henrik Alsing Friberg, and Stefan Ropke (2015). "A note on a model for quay crane scheduling with non-crossing constraints". In: **Engineering Optimization** 47.6, pp. 860–865. DOI: 10.1080/0305215X.2014.958731

Publications Submitted to journals

2026

26. Alberto Santini (2026). "Solving the Berth Allocation Problem via Selective Graph Coloring". Submitted, first round of revision in: **TOP — Transactions in Operations Research**

Technical Reports

2015

27. Valentina Cacchiani, Alberto Santini, and Daniele Vigo (2015). *A study on metaheuristic approaches for the energy-efficient Train Timetabling Problem*. Tech. rep. OR-15-6. DEI, University of Bologna

2014

28. Valentina Cacchiani, Alberto Santini, and Daniele Vigo (2014). *A novel modelling approach for the Train Timetabling Problem*. Tech. rep. OR-14-27. DEI, University of Bologna

Conference Talks

2026

29. Alberto Santini, Enrico Malaguti, and Paolo Paronuzzi (2026). "The 0–1 Knapsack Problem with Group Fairness". In: ROADEF (Feb. 24–26, 2026). Tours, France
30. "A park-and-loop routing problem with parking selection and multiple deliverymen" (2026). In: ECCO XXXIX – CO 2026 (May 14–16, 2026). Lisboa, Portugal
31. Alberto Santini, Enrico Malaguti, and Paolo Paronuzzi (2026). "The 0–1 Knapsack Problem with Group Fairness". In: LION20 (June 15–19, 2026). Milan, Italy
32. "A park-and-loop routing problem with parking selection and multiple deliverymen" (2026). In: IFORS (July 12–17, 2026). Vienna, Austria
33. "A park-and-loop routing problem with parking selection and multiple deliverymen" (2026). In: SEIO 2026 (Sept. 2–5, 2026). Santiago de Compostela, Spain

2025

34. Alberto Santini, Minakshi Punam Mandal, and Claudia Archetti (2025). "Tactical staffing and workforce scheduling decisions for green last-mile delivery". In: SEIO 2025 (June 10–13, 2025). Lleida, Spain
35. Alberto Santini and Vikrant Vaze (2025). "Integrated Schedule Planning for Regional Airlines Using Column Generation". In: Sixth Symposium on Aviation Research (Mar. 20–21, 2025). Bari, Italy
36. Alberto Santini, Enrico Malaguti, and Paolo Paronuzzi (2025). "The 0–1 Knapsack Problem with Group Fairness". In: Optimization and Decision Sciences (Sept. 1–4, 2025). Milan, Italy

2024

37. Alberto Santini and Vikrant Vaze (2024). "Integrated Schedule Planning for Regional Airlines Using Column Generation". In: Annual Workshop of the Transportation Group of the Spanish Society of Statistics and Operational Research (Nov. 28–29, 2024). Seville, Spain
38. Alberto Santini and Vikrant Vaze (2024). "Integrated Schedule Planning for Regional Airlines Using Column Generation". In: Optimization and Decision Sciences (Sept. 8–12, 2024). Badesi, Italy
39. Alberto Santini and Vikrant Vaze (2024). "Integrated Schedule Planning for Regional Airlines Using Column Generation". In: 33rd European Conference on Operational Research (June 30–July 3, 2024). Copenhagen, Denmark
40. Alberto Santini, Minakshi Punam Mandal, and Claudia Archetti (2024). "Tactical staffing and workforce scheduling decisions for green last-mile delivery". In: Odysseus Conference (May 19–24, 2024). Carmona, Spain

2023

41. Alberto Santini and Enrico Malaguti (2023). "The min-Knapsack Problem with Compactness Constraints". In: 19th Meeting of the EURO Working Group on Cutting and Packing (May 3–5, 2023). Bologna, Italy

42. Alberto Santini, Diego Delle Donne, and Claudia Archetti (2023). "Sustainable last-mile delivery using public transport". In: 6th Conference of Young Researchers of the Royal Spanish Society of Mathematics and Statistics (Feb. 6–10, 2023). León, Spain

2022

43. Alberto Santini and Claudia Archetti (2022). "The hazardous orienteering problem". In: 32nd European Conference on Operational Research (July 3–6, 2022). Helsinki, Finland
44. Alberto Santini and Claudia Archetti (2022). "The hazardous orienteering problem". In: SEIO 2022 (June 7–10, 2022). Granada, Spain
45. Alberto Santini, Michael Schneider, Thibaut Vidal, and Daniele Vigo (2022). "Decomposition strategies for vehicle routing heuristics". In: Route 2022 (**Invitation-only conference**) (May 22–25, 2022). Helsingor, Denmark

2021

46. Alberto Santini and Maxence Delorme (2021). "Energy efficient automated vertical farms". In: Optimization and Decision Sciences (Sept. 14–17, 2021). Rome, Italy
47. Alberto Santini, Michele Monaci, and Ciara Pike-Burke (2021). "The 0–1 Time-bomb Knapsack Problem". In: 31st European Conference on Operational Research (July 11–14, 2021). Athens, Greece

2020

48. Alberto Santini, Ana Viana, Xenia Klimentova, and João Pedro Pedroso (2020). "Machine learning and heuristic approaches to the Probabilistic TSP with Crowdsourcing". In: 4th AIROYoung Workshop (Feb. 5–7, 2020). Bolzano, Italy
49. Alberto Santini and Ana Viana (2020). "Crowdsourcing in last-mile delivery: machine learning and heuristic approaches to the Stochastic TSP with Crowdsourcing". In: 5th Conference of Young Researchers of the Royal Spanish Society of Mathematics and Statistics (Jan. 27–31, 2020). Castelló, Spain

2018

50. Alberto Santini (2018). "A hybrid Machine Learning and ALNS approach to the Orienteering Problem". In: EURO/ALIO International Conference on Applied Combinatorial Optimization (June 25–27, 2018). Bologna, Italy
51. Alberto Santini (2018). "An ALNS heuristic for the Orienteering Problem". In: 29th European Conference on Operational Research (July 8–11, 2018). Valencia, Spain
52. Alberto Santini and Christian Blum (2018). "Solving a class of longest subsequence problems via maximum cliques". In: Optimization and Decision Sciences (Sept. 10–13, 2018). Taormina, Italy

2017

53. Alberto Santini, Fabio Furini, and Enrico Malaguti (2017). "Exact and heuristic algorithms for the partition colouring problem". In: Optimization and Decision Sciences (Sept. 4–7, 2017). Sorrento, Italy

2016

54. Alberto Santini, Stefan Ropke, and Lars Magnus Hvattum (2016). "A comparison of acceptance criteria for the Adaptive Large Neighbourhood Search metaheuristic". In: 28th European Conference on Operational Research (July 3–6, 2016). Poznań, Poland
55. Alberto Santini, Stefan Ropke, and Lars Magnus Hvattum (2016). "Poster: A comparison of acceptance criteria for the ALNS metaheuristic". In: 46th Annual Conference of the Italian Operational Research Society (Sept. 6–9, 2016). Trieste, Italy

2015

56. Alberto Santini, Stefan Ropke, and Christian E.M. Plum (2015). "A rich maritime extension of the Vehicle Routing Problem". In: 45th Annual Conference of the Italian Operational Research Society (Sept. 7–10, 2015). Pisa, Italy
57. Alberto Santini, Enrico Malaguti, and Silvano Martello (2015). "A maritime version of the Travelling Salesman Problem". In: Odysseus Conference (May 31–June 5, 2015). Ajaccio, France

2014

58. Alberto Santini, Enrico Malaguti, and Silvano Martello (2014). "The Travelling Salesman Problem with pickup, delivery and draught limits". In: 44th Annual Conference of the Italian Operational Research Society (Sept. 2–5, 2014). Como, Italy

Awards and distinctions

2025 — SSRN Top Paper
The preprint of "Integrated Schedule Planning for Regional Airlines Using Column Generation" was the top paper in SSRN for category "Transport" during Feb–Apr 2025.

2025, 2024, 2023, 2022 — Transportation Science meritorious service award
To recognise "Associate Editors and reviewers who have offered exceptional service to the journal in the review process" during the year 2024.

2023 — INFORMS Journal on Computing featured article
The paper "Decomposition strategies for vehicle routing heuristics" was a featured article in IJOC Volume 35 Issue 3.

2018 — Omega best paper award
Won the Best Paper Award for a paper published in 2018 in the journal Omega. The paper was "The Traveling Salesman Problem with Pickup, Deliveries, and Draft Limits".

Grants Personal or as P.I., competitive

2025–2028 — Single-PI "Generación de Conocimiento" Grant
Three-year grant by the Spanish Government. Project-title: "Fairness and social well-being in optimisation models" (Equidad y bienestar social en los modelos de optimización). Amount: 134,200€.

2024–2029 — Ramón y Cajal Fellowship
Five-year grant by the Spanish Government. Ranked 4th in the Mathemathics panel, with a score of 96.5/100.

2021–2023 — Marie Skłodowska-Curie Fellowship
Marie Skłodowska-Curie Action (MSCA grant number 945380, Cofund), with a project titled "SILD: Sustainable Integrated Last-mile Delivery". Grant to join ESSEC Business School.

2021 — Juan de la Cierva Incorporación
Three-year grant by the Spanish government. *Refused* because it was incompatible with the Marie Skłodowska-Curie fellowship. Ranked 5th out of 28 participants from all areas of Mathematics, with a score of 90/100.

2021 — BGSE Seed Grant
Grant awarded by the Barcelona School of Economics; topic: "Sustainable Last-mile Delivery". Amount: 10,000€.

2020–2021 — Planetary Wellbeing research grant
PI of a research proposal with Gara Villalba (Universitat Autonoma de Barcelona) and Michael Schneider (RWTH Aachen University) on the optimisation of vertical farming in Barcelona's public schools. Grant awarded by Universitat Pompeu Fabra. Amount: 17,880€.

2019–2021 — Juan de la Cierva Formación
Two-year grant by the Spanish government. Ranked 1st out of 49 participants from all areas of Mathematics, with a score of 99/100.

2014–2016 — Ph.D. grant by the University of Bologna
Ranked 1st out of 9 participants, with a score of 92/100.

Grants As part of a group, competitive

2018–2021 I+D+i 2018 (Retos)
Part of the project "Optimización logística para el cuidado de la salud en el hogar y los servicios sociales" funded by the Spanish Ministry of Science. Amount: 35,453€.

2017–2020 PRIN (Research Projects of Relevant National Interest) 2015
Part of the project "Transportation and Logistics Optimization in the Era of Big and Open Data" funded by the Italian Ministr of Research.

Grants Other

2021 Renowned Guest Visitor Grant
Granted by the Research Centre of ESSEC Business School.

2019 Yerun Mobility Award
To visit University Paris Dauphine, sponsored by YERUN.

2018–2023 Association of European Operational Research Societies (EURO)
To organise the EUROYoung Workshops.

2018 Erasmus+ Grant
To visit the University of Bologna, through the SmartTransLog consortium.

2017 COST STSM Grant
Short-Term Scientific Mission grant to conduct research at INESC-TEC under COST Action "European Network for Collaboration on Kidney Exchange Programmes".

2016 Association of European Operational Research Societies (EURO)
To organise the "EURO Young Researchers OR Webinars".

2015 Marco Polo scholarship
To visit the Technical University of Denmark for a 3-month research stay.

Teaching Ph.D. level

2019–2020 Business Analytics for Operations and Marketing
Taught in a team. Universitat Pompeu Fabra, first-year course for Ph.D. students (M.Res. in Economics, Finance and Business).

2018–2021 Research Seminars
Taught in a team. Universitat Pompeu Fabra, first-year course for Ph.D. students (M.Res. in Economics, Finance and Business).

Teaching Master's level

2026 From Prediction to Action: Modern Prescriptive Analytics
Barcelona School of Management, Universitat Pompeu Fabra, M.Sc. in Management.

2025 Foundations of Data Science
Taught together with Lorenzo Cappello. Barcelona School of Economics, M.Sc. in Economics and Finance, Financial Economics Programme.

2022–2025 Deterministic Models and Optimisation
Barcelona School of Economics, M.Sc. in Data Science, Data Science Methodology Programme.

2021–2022 Deterministic Models and Optimisation
Taught together with Jessica Rodríguez-Pereira. Barcelona School of Economics, M.Sc. in Data Science, Data Science Methodology Programme.

2021–2025 Practical Data Science for Finance
Barcelona School of Management, Universitat Pompeu Fabra, M.Sc. in Finance and Banking.

2019–2022 Practical Data Science for Management
Barcelona School of Management, Universitat Pompeu Fabra, M.Sc. in Management.

2019–2023 Advanced Machine Learning
RWTH Business School, M.Sc. in Data Analytics and Decision Science.

2018–2019 Data Science Projects
RWTH Business School, M.Sc. in Data Analytics and Decision Science.

2018–2023 Machine Learning
RWTH Business School, M.Sc. in Data Analytics and Decision Science.

2017 Optimisation methods for Big Data Analytics
RWTH Aachen, Master's in Management, Business, and Economics.

2014–2018 Optimisation Algorithms (teaching assistant)
Course taught by Paolo Toth.
University of Bologna, M.Eng. in Computer Engineering.

Teaching Bachelor's level

2025 From Prediction to Action: Modern Prescriptive Analytics
Universitat Pompeu Fabra, Barcelona Programme for Interdisciplinary Studies (BaPIS).

2018–2021, 2024–2025 Operations Management
Universitat Pompeu Fabra, Bachelor's in International Business Economics.

2018–2019, 2024 Operations Management
Universitat Pompeu Fabra, Bachelor's in Business Management Administration.

2021 Applied Machine Learning and Optimisation
Universitat Pompeu Fabra, elective course for all last-year Bachelor's students of the Department of Economics.

2020 Applied Optimisation Techniques
Universitat Pompeu Fabra, elective course for all last-year Bachelor's students of the Department of Economics.

2019 Revenue Management and Dynamic Pricing
Universitat Pompeu Fabra, elective course for all last-year Bachelor's students of the Department of Economics.

2016–2017 Foundations of OR (teaching assistant)
Course taught by Enrico Malaguti and Michele Monaci.
University of Bologna, B.Eng. in Management Engineering.

Academic experience Scientific Organisations

2024–now Coordinator
Coordinator of the Working Group on Transportation of the Spanish Statistics and Operational Research Society.

2018–2024 Coordinator
Coordinated the activities of EUROYoung, including four international workshops, a Ph.D. school, and sessions and round tables at the EURO conferences.

2019–2021 Elected member of the Board of Directors
Member of the board of directors of AIRO, the Italian Operational Research Association.

2018 — Founder
Founded EUROYoung, a group of young O.R. scientists from EURO member countries, with the aim of organising training events for early-stage researchers.

2016–2018 — Coordinator
Coordinator of AIROYoung, the youth chapter of the Italian Operational Research Association. During my tenure as coordinator, I organised two workshops securing funding from Italian and European institutions; I created a database of young talents in O.R. and of open positions, helping to match O.R. scientists with relevant opportunities; I helped enrol new young members into AIRO, from Italy and other European countries.

2016 — Founder
Founder of AIROYoung, the youth chapter of the Italian Operational Research Association.

Academic experience Organisation of Seminars, Workshops, and Courses

2025 — Organiser
The 1st PhD School and the annual workshop of the Working Group on Transportation of the Spanish Statistics and Operational Research Society. Invited speaker: Roberto Roberti. Barcelona, Spain, 1–3 December 2025.

2024 — Member of the Organising Committee
The 1st EUROYoung PhD School. Invited speakers: Michel Bierlaire, David Pisinger, Claudia Archetti, Elena Fernandez and Ola Jabali. Barcelona, Spain, 4–6 September 2024.

2023 — Member of the Organising Committee
The 3rd EUROYoung Workshop. Invited speakers: Ivana Ljubić and Leo Liberti. Cergy (Paris), France, 5–6 June 2023.

2022 — Member of the Organising Committee
The 2nd EUROYoung Workshop. Invited speakers: Renata Mansini and Pedro Amorim. Porto, Portugal, 25–26 June 2022.

2019 — Organiser of a PhD and Postdoc course
Organised a course for PhD and Postdocs of the Barcelona Graduate School of Mathematics, titled "An Introduction to Equilibrium Problems and their applications". The course is taught by Mauro Passacantando.

2019 — Member of the Organising Committee
The 1st EUROYoung Workshop. Invited speakers: Ana Viana and Federico Perea. Seville, Spain, 2–3 May 2019.

2019 — Member of the Organising Committee
The 3rd AIROYoung Workshop and 1st AIROYoung PhD School. Invited speakers: Teodor Crainic and Endre Boros. Rome, Italy, 26–29 Mar 2019.

2018–2019 — Coordinator of the "Operations and Marketing" seminar series
This is a series of weekly seminars at the Department of Economics and Business of Universitat Pompeu Fabra, with invited speakers from all over the world.

2018 — Member of the Organising Committee
The 2nd AIROYoung Workshop, on the theme "Logistics optimisation and applied operational research". Invited speakers: Geir Hasle and Roberto Roberti. Cosenza, Italy, 03–05 Mar 2018.

2017 — Workshop Organiser
Organised, together with Lavinia Amorosi, the 1st AIROYoung Workshop, on the theme "Emerging Optimization Problems on Complex Networks". Invited speakers: Justo Puerto and Maria Grazia Scutellà. Rome, Italy, 16–17 Feb 2017.

2016 — EURO Young Researchers OR Webinars Organiser

Organised a series of OR webinars by and for young researchers from all over the world, sponsored by EURO.

2015 Seminars Organiser
Organised of a series of doctoral seminars within the Operational Research group at DEI, University of Bologna, involving local and visiting researchers.

Academic experience Organisation of Conferences

2027 Member of the Programme Committee
EURO 2027 Conference. Athens, Greece, 11–14 July 2027.

2026 Member of the Programme Committee
LOGMS (Logistics and Maritime Systems) 2026 Conference. Barcelona, Spain, 7–10 July 2026.

2025 Session Chair
"Logistics and Last-Mile Delivery" session at the SEIO 2025 Conference. Lleida, Spain, 10–13 June 2025.

2024 Member of the Programme Committee
ISCO2024, the International Symposium on Combinatorial Optimization. Tenerife, Spain, 22–24 May 2024.

2023 Member of the Programme Committee (ECOM track)
GECCO2023, the Genetic and Evolutionary Computation Conference. Lisbon, Portugal, 15–19 July 2023.

2022 Session Chair
"MILP in Logistics and Last-Mile Delivery" session at the EURO 2022 Conference. Espoo, Finland, 3–6 July 2022.

2022 Session Chair
"Recent advances in transportation and supply chain optimisation" session at the SEIO 2022 Conference. Granada, Spain, 07–10 June 2022.

2022 Member of the Programme Committee
ISCO2022, the International Symposium on Combinatorial Optimization. Online, 18–20 May 2022.

2022 Member of the Programme Committee (ECOM track)
GECCO2022, the Genetic and Evolutionary Computation Conference. Boston, United States, 9–13 July 2022.

2022 Member of the Programme Committee
ODS2022, Optimisation and Decision Science 2022. Florence, Italy, 30 August – 02 September 2022.

2021 Member of the Programme Committee (ECOM track)
GECCO2021, the Genetic and Evolutionary Computation Conference. Lille, France, 10–14 July 2021.

2020 Member of the Programme Committee
ODS2020, Optimisation and Decision Science 2020 with theme "Optimisation in Artificial Intelligence and Data Science". Rome, Italy, 19 November 2020.

2020 Member of the Programme Committee (ECOM track)
GECCO2020, the Genetic and Evolutionary Computation Conference. Cancún, Mexico, 08–12 July 2020.

2019 Member of the Programme Committee (ECOM track)
GECCO2019, the Genetic and Evolutionary Computation Conference. Prague, Czech Republic, 13–17 July 2019.

2019 Member of the Programme Committee
HM2019, the 11th International Workshop on Hybrid Metaheuristics. Concepción, Chile, 16–18 January 2019.

2018 Member of the Organising Committee and Session Chair
ODS 2018, Optimisation and Decision Science Conference. Chair of the "AIROYoung" session. Taormina, Italy, 10–13 September 2018.

2018 Session Chair
"EURO Young" session at the EURO 2018 Conference. Valencia, Spain, 08–11 July 2018.

2018 Session Chair
"AIROYoung" session at the EURO/ALIO 2018 Conference. Bologna, Italy, 25–27 June 2018.

2017 Session Chair
"AIROYoung" session at the ODS 2017, Optimisation and Decision Science Conference. Sorrento, Italy, 04–07 September 2017.

2016 Poster Session Chair
"Young OR Researchers Poster Session" at the AIRO 2016 Conference. Trieste, Italy, 06–09 September 2016.

2016 Session Chair
"Methodological advancements in metaheuristics" session at the EURO 2016 Conference. Poznan, Poland, 03–06 July 2016.

2015 Session Chair
"Exact Methods for Routing Problems" at the AIRO 2015 Conference. Pisa, Italy, 07–10 September 2015.

Academic experience Evaluation Committees and Reviews

2026 Member of the evaluation committee for the "Generación de Conocimiento" grants
Agencia Estatal de Investigación, Spain.

2026 Member of the evaluation committee for the "Juan de la Cierva" and "Ramón y Cajal" scholarships
Agencia Estatal de Investigación, Spain.

2023, 2024 Member of the Best PhD Dissertation Award committee
AIRO, the Italian Operational Research Society.

2020–now Member of PhD defence committees
At the University of Modena and Reggio Emilia (Italy), the University of Palermo (Italy), the Open University of Catalonia (Spain), Universitat Pompeu Fabra (Spain), Universitat Politecnica de Catalunya (Spain).

2018 Grant proposal evaluator
"FONDECYT Initiation into Research" programme, Chilean National Science and Technology Commission.

2018–2020 Member of the permanent M.Sc. thesis defence committee
Master's in Statistics and Operational Research, Polytechnic University of Catalunya, Spain.
Master's in Management, Barcelona School of Management, Spain.

2014–present Reviewer for several international journals, including: Operations Research, Transportation Science, INFORMS Journal on Computing, European Journal of Operational Research, Computers & Operations Research, Omega, Transportation Research: Part B, Transportation Research: Part C, Transportation Research: Part E, International Transactions in Operations Research, Networks, OR Spectrum, Operations Research Forum, Discrete Applied Mathematics, Optimization Letters, Transportation Letters, Informations Systems and Operational Research, Operations Research Perspectives, Expert Systems With Applications, IEEE Access, Journal of Heuristics, Applied Mathematical Mod-

elling.

Academic experience Visiting Positions

2018 University of Bologna, Italy
Visiting Paolo Toth and Daniele Vigo under an Erasmus+ grant from the SmartTransLog consortium.

2017 INSEC-TEC, Portugal
Visiting as part of a Short-Term Scientific Mission for COST Action "European Network for Collaboration on Kidney Exchange Programmes". Working with Ana Viana, João Pedro Pedroso, and Xenia Klimentova.

2016 Amazon Inc.
Visiting Research Scientist in Seattle, USA. Working with Mauricio Resende and Andrew Goldberg.

2016 Vrije Universiteit Amsterdam
Visiting Ph.D. Student at the Department of Information, Logistics and Innovation. Working with Wout Dullaert.

2015 University of Calabria
Visiting Ph.D. Student at the Department of Mechanical, Energy and Management Engineering. Working with Demetrio Laganà.

2015 Danish Technical University (DTU)
Visiting Ph.D. Student at the Department of Management Engineering. Working with Stefan Røpke.

Invited talks

2026 "Decomposition methods for large-scale vehicle routing problems"
Invited talk at the "Laboratoire d'Informatique, de Robotique et de Microélectronique de Montpellier", France. Montpellier, 5 May 2026.

2024 "Using public transport to deliver parcels in large cities"
Invited talk at the Department of Economics and Management of the University of Brescia, Italy. Brescia, 7 November 2024.

2024 "Shift stability and workers' wellbeing in last-mile delivery scheduling"
Internal Operational Research seminar series at the Department of Economics and Business, Universitat Pompeu Fabra. Barcelona, 2 July 2024.

2023 "Two non-linear stochastic problems with catastrophic consequences"
Invited seminar at the Laboratoire d'Informatique de Paris Nord. Paris, 22 March 2023.

2022 "Decomposition methods for vehicle routing heuristics"
Invited to present at the 3rd Spanish Young Statisticians and Operational Researchers Meeting. Elche, Spain, 21–23 September 2022.

2022 "Blending ML and OR tools to solve a problem in crowdsourced grocery home delivery"
Invited seminar at the Tilburg School of Economics and Management. Tilburg, the Netherlands, 7 July 2022.

2022 "Delivering parcels, growing lettuce on Mars and transporting time-bombs using one single tool"
Guest lecture at the Institute of Advanced Studies. Neuville, France, 15 March 2022.

2021 "The 0–1 Time-bomb Knapsack Problem"
Invited to the "Research Seminars" series at the Department of Information Systems, Decision Sciences and Statistics, ESSEC. Cergy, France, 15 December 2021.

2021 "New challenges in supply chain optimisation"
Invited to the OR-OM Seminar Series at ESSEC. Cergy, France, 1 December 2021.

2021 "The importance of networking in career development"
Invited to the WISDOM round table within the EURO 2021 conference.

2019 "Crowdsourcing in last-mile delivery: machine learning and heuristic approaches to the Stochastic TSP with Crowdsourcing."
Invited to present at the seminar series of LAMSADE, University Paris Dauphine. Paris, France, 26 November 2019.

2016 "A real-time conflict resolution algorithm for the train rescheduling problem."
Invited to present at: Vrije Universiteit, Amsterdam (Feb 2016); Danish Technical University (DTU), Kongens Lyngby (Oct 2016); DSB Training Centre, Nyborg (Oct 2016).

2015 "TSP with pickups, deliveries, and draught limits."
University of Bologna. Part of a series of seminars held by Ph.D. students and young researchers.

2014 "Data Enevelopment Analysis and OR applications in measuring efficiency."
University of Bologna. Part of the course "Resource Optimization", M.Eng. in Management Engineering.

Outreach activities

2022 "Il senso della scienza ovvero la bellezza delle scienze."
Workshop with high-school students. Liceo Scientifico Statale "Alessandro Volta", Caltanissetta, Italy, 24 November 2022.

2014 "Recent trends in Operational Research and practical business applications."
American University of Armenia; series of two seminars at the College of Science and Engineering. Yerevan, Armenia, 9–10 July 2014.

Supervision Ph.D. students

2026–2030 Ph.D. Programme in Information and Communication Technologies
Supervisor of Amit Entin. Thesis title: TBD.

2026–2031 Ph.D. Programme in Economics, Finance and Management
Supervisor of Anton Zudin. Thesis title: TBD.

Supervision Visiting students

2026–2027 Visiting Ph.D. student
Zhehan Liu from the National University of Defense Technology, Changsha, China. Visit topic: Tracking, telemetry and control scheduling for large-scale low earth orbit internet constellations.

2026 Visiting Ph.D. student
Lorenzo Saccucci from the University of Rome "La Sapienza". Visit topic: exact methods for a vehicle routing problem with on-time arrivals.

2019 Visiting Erasmus+ intern
Vinicius Greco de Lemos from the University of Twente. Visit topic: production planning and optimisation of vertical farms.

Supervision Bachelor's and Master's theses

2025 Bachelor's Thesis, B.Sc. in International Business Economics, Universitat Pompeu Fabra
Supervisor, "Forecasting wealth and anticipating tax", Quim Roselló, Oriol Diaz.

2024 Master's Thesis, Master's Programme in Data Science Methodology, Barcelona School of Economics
Supervisor, "Putting workers' wellbeing at the centre of rostering decisions in last-mile delivery and retail", Jonny Codd, Clarice Mottet, Carlos Pérez.

Master's Thesis, Master's Programme in Data Science Methodology, Barcelona School of Economics
Supervisor, "A branch-and-price algorithm for a rich vehicle routing problem with synchronisation", Xi Cheng, Yutai Ke.

2023 Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Understanding the propagation of glass defects in windshields before and after lamination using high-resolution geometric measurement data", Ju-Chin Hsiao.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Improving the performance of a deep learning industrial part recognition system", Sumedh Sunil Thanekar.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Portfolio optimization using the 'Smart Predict then Optimize' framework", Tejas Govind Choudekar.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Estimation of the resale value of electric cars using machine learning and interpretable models", Jeswin Aloysius.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Neural networks for disease end-point prediction", Karan Sanan.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Prediction of tyre uniformity for truck tyres using machine learning", Wei-Chi Huang.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Predicting purchase requisition processing time using machine learning", Khartikeyan Ramasubbu.

2022 Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Uncertainty modelling in industrial production processes: predicting aleatoric and epistemic uncertainty for quality assessment of nonwovens", Himanshu Patanwala.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Unsupervised anomaly detection in multivariate time series", Kevin Meda.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Predicting ghosting effect of HUD systems before wind-shield lamination", Ju-Chin Hsiao.

2021 Master's Thesis, M.Sc. in Management, Barcelona School of Management, Universitat Pompeu Fabra
Supervisor, "Micro-economic insights for realistic pricing in the Vehicle Routing Problem with Occasional Drivers", Helena Bonnin.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Generalisability of machine learning algorithms for selective assembly", Sourabh Hujare.

Master's Thesis, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Machine learning applied to metamodelling of thermo-structural systems and their functional mock-up interface", Clinton Charles.

2020 Master's Thesis, M.Res. in Economics, School of Economics and Business, Universitat Pompeu Fabra
Co-supervisor, "Imitation, Principal Component Analysis, and the Effects of Modifications to eSports

on Competitive Balance", Karim Carroum.

Master's Thesis, M.Sc. in Management, Barcelona School of Management, Universitat Pompeu Fabra
Supervisor, "A simulation study of integrated real-time data resulting in a dynamic pricing model for crowdsourcing-to-customer last-mile deliveries", Otis van der Meij.

Project, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "COVID-19 Pandemic: Comparison of forecasting methodologies and affecting factors for fatalities and transmission rates", Peter Yacoub.

Project, M.Sc. in Data Analytics and Decision Science, RWTH Business School
Supervisor, "Machine learning for selective assembly with automatic adaption to use-case complexity", Sourabh Hujare.

Bachelor's Thesis, School of Economics and Business, Universitat Pompeu Fabra
Supervisor, "Congestion tax as the optimal policy tool to minimize congestion in the city of Barcelona", Joel Sanginés-Uriarte Dooly and Adrià Trepat i Izquierdo.

Bachelor's Thesis, School of Economics and Business, Universitat Pompeu Fabra
Supervisor, "Machine Learning for Price Discrimination", Jan Czernek.

2019 Master's Thesis, M.Sc. in Management, Barcelona School of Management, Universitat Pompeu Fabra
Supervisor, "Demand forecasting model in food delivery, a heuristic approach to data science using Python and geoanalysis", Marc Herrera.

2017 Project, M.Sc. in Management, Business, and Economics, RWTH Aachen
Supervisor, "Improving document-level machine translation with `word2vec`", Sean Eulenberg.

Master's Thesis, M.Sc. in Management, Business, and Economics, RWTH Aachen
Co-supervisor, "A heuristic algorithm for the electric pickup and delivery problem", Lutz Venhofen.

2016 Master's Thesis, M.Eng. in Management Engineering, University of Bologna
Co-supervisor, "Embedding robustness in train rescheduling algorithms", Alessandro Morandi.

Project, M.Eng. in Computer Engineering, University of Bologna
Co-supervisor, "A genetic algorithm for real-time energy optimisation of train timetables", Francesco Di Lella.

2015 Master's Thesis, M.Eng. in Management Engineering, University of Bologna
Co-supervisor, "Metaheuristic algorithms for energy optimisation in train timetabling and rescheduling", Hang Xu.

Project, M.Eng. in Computer Engineering, University of Bologna
Co-supervisor, "A constructive heuristic for the dispatching problem on multi-track territories", Giacomo Dall'Olio and Simone Laierno.

2014 Master's Thesis, M.Eng. in Management Engineering, University of Bologna
Co-supervisor, "Valutazione della Performance di un caso di Customer Contact Desk attraverso l'applicazione della metodologia Data Envelopment Analysis", Martina Carli.

Work experience

2016 Amazon
Research Scientist Intern. Seattle, USA.

2012–2013 Procter & Gamble
Strategic Sourcing Intern. Bruxelles, Belgium.

2012 WISON Offshore & Marine
Procurement Trainee. Shanghai, China.

2011–2012 Amiiko
Ruby on Rails Developer. Copenhagen, Denmark.

Other

Languages	Italian: mothertongue; English: fluent; French: very good; Spanish: very good; Catalan: very good; Armenian: constantly improving!
Volunteering	Spanish Red Cross; Italian Red Cross; Danish Red Cross; Big Brother / Big Sister Project Kongelunden, Copenhagen, Denmark; Associació per la Promoció del Transport Public, Barcelona, Spain.
Certifications	Chartered shipbroker according to Italian law (2012). Ham Radio operator (EA3JEU).